

# Roof-Mounted PV Systems

by Tracy Fox

One of the most important things you can do when designing and installing a PV system is to get to know the customer's roof. While there are many facets to designing and installing solar electric systems, the home site will ultimately determine the type of PV system you install, and where you install it. For most folks that means the rooftop, where often there is more than meets the eye.

For the homeowner, whether or not it makes sense to install a PV system boils down to two things: performance and cost. There are several factors that influence a system's performance. Homeowners need to understand these factors so that they have realistic expectations and remain satisfied with their choice of PV system. Most PV systems nominally produce about 10 watts of power for each square foot of PV module. While smaller starter residential systems require as little as 50 square feet of space, a more powerful system can cover 1,000 square feet of unobstructed rooftop.

External factors, such as the type of PV module, the home's geographic location, and the tilt and orientation of the roof, play a big role in system output. The internal workings of a home also affect net output, and must be considered when sizing the PV system. The energy efficiency of the home's appliances and HVAC system, as well as the home's airtightness and insulation levels, determine the PV system's ability to offset the electricity bill. When customers can see the big picture, they're more likely to work with you to create the best overall system for their home.



BRUCE WILCOX

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## OPTIMUM ORIENTATION

John Wiles, program manager at the Technology Development Institute at New Mexico State University, explains that the optimum orientation for a fixed-array PV system in the Northern hemisphere is true south, not magnetic south. To maximize the output of a PV system, the modules should be tilted to an angle from the horizon equal to the local latitude. Variations to this angle should be

based on four factors. These are (1) the type of system (utility interactive or stand-alone), (2) any net metering restrictions on monthly excess energy carryover, (3) the local microclimate, and (4) array shading. There should be no shading of any type from 9 am to 4 pm on the shortest day of the year. This includes shading from small obstructions such as chimneys and vent pipes, which can have a disproportionately high impact on performance. Leaving room for air to circulate and cool the panels on both sides also improves system output. "Anything that

deviates from an optimum alignment or orientation and optimum cooling is going to decrease your performance,” says Wiles.

PV modules can be expected to last well over 20 years when designed, installed, and maintained properly. Other components of the PV system may not last as long, and may need to be replaced during the life of the modules. It is important to provide access to the modules for maintenance purposes; this can add up to 20% to the total space required for a PV system. There is also the very strong likelihood that the system will outlast the roof, which will probably need to be replaced within 20 years. Therefore, it is important for the homeowner to understand that a PV system may need to be disassembled at some point in the future.

## ROOF MOUNT

Before installing a PV system, you should thoroughly examine the customer’s roof. If the roof doesn’t appear to be in tip-top shape, call in a roofing contractor to repair, or perhaps even to replace, it. Homes built before 1970 should have a structural engineer’s input to determine if the roof can support the PV array; these homes may need structural enhancements.

The most common method of mounting a PV array on a sloped roof is to use standoff mounting racks. These are permanent racks that hold the modules at a fixed-degree angle; some models can be adjusted throughout the year. While some installers design and fabricate their own mounting structures, mounting system hardware purchased from a manufacturer can be engineered to withstand specific wind speeds and can be designed for a specific roof. Manufactured roof mounts generally don’t require independent engineering analyses or additional documentation for building inspectors. The mounting system hardware is also warranted

by the manufacturer, although the details of each warranty will differ. All this gives the advantage to using manufactured roof mounts rather than custom-made ones.

On a flat roof, the PV arrays may be mounted on a separate structure, to provide the best tilt angle, or the arrays may be mounted flat on the roof, facing the sky. Thoroughly sealing all penetrations is a major concern when it comes to roof mounting—for obvious reasons—as nobody wants a callback on a leaking roof.

Alex Aragon has worked in the electrical industry for nine years and has been doing solar installations for the last four. He teaches workshops at the Solar Living Institute in Hopland, California. Aragon emphasizes the importance of talking to the homeowner throughout the entire design and installation process. He recommends identifying the type of roof construction; looking at the site and establishing whether or not the roof is a good candidate for a PV system; and viewing the roof from the perspective of existing warranty provisions. Will the layout of the roof work? If everything is pliable and the shingles aren’t brittle and cracking, talk to the homeowner about a roof mount installation and about any penetrations. If the roof is still under warranty, the warranty may not cover penetrations made by a solar installer. If the roof is brand-new, Aragon recommends calling back the original roofing contractor to install roof jacks and flashing, so as not to void the original warranty. Some roofers may insist on using mounting systems that require common roof flashing to ensure a watertight seal.

If a home has a flat or minimally sloped roof, like the ones that crown Eichler-style homes in California, and if it is necessary to install a new roof, an ideal choice for a roofing material is spray polyurethane foam (SPF). SPF roofing is a monolithic watertight material with an R-value of 6–6.8 per inch, and it weighs a mere 50 lb per 100 square feet of roof at a thickness

of 1 to 1-1/2 inches (aggregate-coated foam adds around 60 lb per 100 square feet). This makes it an attractive choice for a roof on which a PV system is to be installed. Adam Feriante, who is in charge of production at Dura-Foam Roofing, suggests that the solar contractor should deal directly with the foam roofing contractor. “There is more than one way to do a mount and to do it effectively,” says Feriante, “and in some cases, the solar contractor can get specific instructions on how to seal the penetration.” For example, when there are fewer than a dozen penetrations, the solar installer can use galvanized metal roof jacks and a single-component urethane sealant. When there are more than a dozen penetrations, it is best to have the roofing contractor come out with the foam truck. With UV coatings and proper maintenance, an SPF roof can last 30 years or more. As long as the foam is protected from UV degradation, it doesn’t break down, and when it comes time to recoat an SPF roof, the coating can be sprayed around the perimeter of the panels without having to disassemble them.

## STRUCTURAL LOADS

Underneath all the roofing materials are the trusses—the Hercules of the home—and for sloped roofs, the mounts must be fastened either to the roof trusses or to the rafters, not to the roof decking. Truss manufacturers’ load-bearing calculations are based both on local building codes and on the type of roof—on whether it is asphalt shingle or concrete tile roofing, for example. However, truss manufacturers don’t automatically include the weight of a PV system or wind loads in their calculations, and these weights can be large enough to affect structural integrity. Weight can add up quickly on a rooftop, especially when more than one layer of shingles is on the roof. Composition shingle weighs 200–400 lb per 100 square feet, and concrete and clay tiles tip the





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scales at 500–1,200 lb per 100 square feet. Masonry roofs are often structurally designed near the limit of their weight-bearing capacity. Changing out the masonry roofing to a lighter roofing, such as composition shingle, in the area where the PV array is to be installed eliminates the need to reinforce the roof, because the combined weight of the PV array and the shingle is less than the weight of the masonry roofing alone.

PV modules, including the racks, add about 400 lb per 100 square feet to the roof's structural load. The strain on the roof increases as the wind interacts with the PV array—in a strong wind, total uplift force may reach 50 lb per 100 square feet. The Wood Truss Council of America recommends installing roof diaphragms to accommodate wind load under extreme conditions. The North American Board of Certified Energy Practitioners (NABCEP) recommends keeping the modules away from the edges of the roof in regions where wind load may be significant.

## BUILDING INTEGRATED PV

A building integrated PV (BIPV) roofing system performs the functions of a roof while generating electricity (see “Designing for Solar,” p. 18). BIPV products include roofing tiles, shingles, slates, and standing seam metal roofing. BIPV currently represent less than 5% of solar-electric installations nationwide, but they are fast growing in popularity. In California, where over 85% of all new U.S. solar-powered homes are built, BP Solar and Old Country Roofing will team up to offer builders and homeowners a turnkey solar-roofing package in 2007 that will include design, installation, and a warranty. Homeowners like BIPV for both new construction and reroofing, both for its two-for-one appeal, and because it integrates seamlessly with most roofing materials.

The economic value of installing a PV system varies by region—that is, by the amount of available sunlight—and by the size and features of the system itself. The cost of these systems can be offset by state incentives, tax credits, the increase in a home's resale value, and lower utility bills. Over its

projected lifetime, the noiseless, clean energy production provided by a PV system is enough to make anyone feel good—really good—about the decision to install a PV system.



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